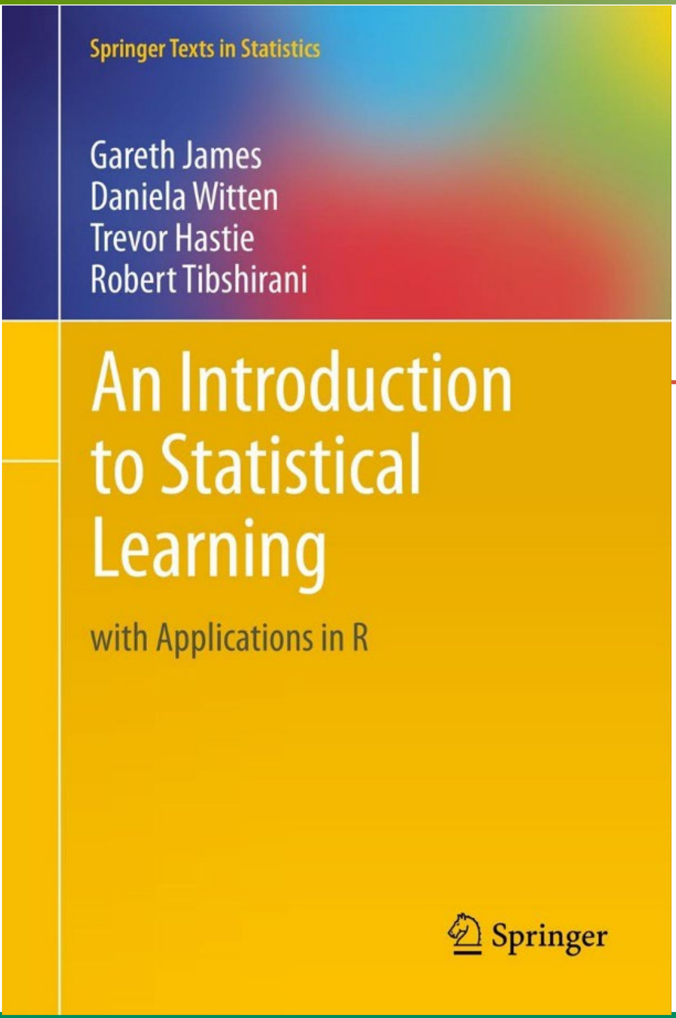

ANALYTICS

Week 3-4

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Examples Inspired By CAS, SOA, Coaching Actuaries ©

Key Formulas This Week

- Residual standard error: $s = RSE = \sqrt{MSE} = \sqrt{\frac{RSS}{n-2}}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$, $se(b_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- $se(\hat{y}) = s\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- $se(\hat{y}_{pred}) = s\sqrt{1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{(n-1)s_x^2}} = s\sqrt{1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- Confidence Interval: $b_1 \pm t_{n-2,\frac{\alpha}{2}} * se(b_1)$
- Prediction Interval: $\hat{y}_{pred} \pm t_{n-2,\frac{\alpha}{2}} * se(\hat{y}_{pred})$

Hypothesis Testing

- Hypothesis testing helps us compare 2 models statistically.
- H_0: The simpler baseline model typically
- H_a: More complex model
- Goal: Is the additional complexity in H_a significant or ____
- How: Calculate a test statistic and compare to a threshold to see if we should reject H_0.

Hypothesis Testing 2

- For basic linear regression $Y = \beta_0 + \beta_1 X + \varepsilon$,
- $H_0: \beta_1 = 0$ (no longer includes the x variable)
- H_a: ____

t-statistic

- Use the t-statistic to help determine significance levels of the hypothesis test.

- t-ratio =

- For testing $H_0\colon \beta_1 = 0$ in basic linear regression,
- $t = \frac{b_1}{se(b_1)}$ with $df = n - 2$

Estimator Variance *

- The uncertainty around the estimated values $b_0, b_1, \hat{y}$ can be quantified using their respective standard errors.
- $Var(b_0) = MSE\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $Var(b_1) = MSE\left(\frac{1}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $Var(\hat{y}) = MSE\left(\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $Var(\hat{y}_{pred}) = MSE\left(1 + \frac{1}{n} + \frac{\left(x_{pred} - \bar{x}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$

Standard Errors

- Standard error is the ___

- More observations (larger n), smaller residual standard deviation $s = \sqrt{MSE}$, and more leverage (higher s_x) will reduce se.
- $se(\hat{y}_{pred})$ is typically much larger due to the 2 sources of variability:
- 1. Uncertainty in estimating the regression line
- 2. ___

1. Var Illustration (SOA)

- Given a SLR model: $Y = \beta_0 + \beta_1 x + \varepsilon$ with 22 observations, the sample correlation coefficient is 0.5, and the estimated slope coefficient is 2.
- Determine the variance of the slope coefficient.

1. Var Illustration 2

- $Var(b_1) = MSE\left(\frac{1}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $MSE = \frac{SSE}{n-2}$
- $R^2 = 1 - \frac{SSE}{SST} = \frac{SSReg}{SST}$
- $SSReg = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = b_1^2 \sum_{i=1}^{n}(x_i - \bar{x})^2$

Example 1 (SOA)

- You use a simple linear regression model to predict Minecoins earned using # of hours played as a predictor, based on 9 observations.
- $b_1 = \sqrt{20}$
- $\sum x_i = 27$
- $\sum x_i^2 = 91$
- $\sum (y_i - \hat{y}_i)^2 = 210$
- Calculate the t-statistic used for testing the hypothesis $H_0: \beta_1 = 1$.

Example 1 Sol

- $MSE = s^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{SSE}{n-2}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}} = \frac{s}{\sqrt{\sum_{i=1}^{n} x_i^2-n\bar{x}^2}}$
- t-ratio $= \frac{Estimator-H_0\ value}{se(Estimator)}$

Var-Cov Matrix

- The variance-covariance matrix can be calculated to find standard errors.

- $Var(b) = \begin{bmatrix} Var(b_0) & Cov(b_0, b_1) \\ Cov(b_0, b_1) & Var(b_1) \end{bmatrix} = MSE(\boldsymbol{X}^T\boldsymbol{X})^{-1}$

Degrees of Freedom

- Degrees of freedom reflects a model's flexibility and the # of ___
- In a simple linear regression, if you have $n - 2$ points and the linear equation, you know the last 2 points must be.
- More flexible models have ___

Test Statistic Decision

- Compare the calculated t to a threshold value (from t-Table) at the chosen significance α.
- Smaller α = ___
- Reject H_0 if $|t| > t_{critical}$ for a 2-sided test, meaning $H_a: \beta_1 \neq 0$
- $t_{critical}$ comes from the t-Table

t-Table

- In SRM, the t-Table shows 1-sided α, so for 2-sided tests, double the Table's α.
- In MAS-I, the t-Table is 2-sided.

2-sided alpha	20%	10%	5%	2%	1%
df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554

p-value

- The p-value is the probability of observing the data sample or a ___

- If $p \leq \alpha$ or $|t| > t_{critical}$, we reject H_0.
- Rejecting H_0 does <u>not</u> mean our estimated b_1 is correct (typically never equal to the true), but rather that it ___

Hypothesis Testing Summary

- Set up H_0 & H_a. Decide on α.
- H_0 is a simpler model and H_a more complex.
- Use the estimated slope coefficient to determine your t statistic.
- Use the t-Table with $df = n - 2$ to determine critical t values at various α. Double the Table α for 2-sided tests.
- If the absolute value of the t statistic > critical t value at α, reject H_0 for that significance level and conclude parameter is useful to include.
- Smaller p-value or larger t = evidence to reject H_0

Test Relationships *

- Small p → Higher t-statistic, more evidence to reject H_0 → Keep parameter (it's significant)

- Large p → ___

- The pre-set standard for rejecting H_0 is α.

F-statistic

- In basic linear regression, the F-statistic is used to test whether the single predictor is useful.
- $H_0: \beta_1 = 0$ vs. $H_a: \beta_1 \neq 0$
- $F = \frac{MSReg}{MSE} = (n-2)\frac{R^2}{1-R^2}$
- For basic linear regression, it is directly related to the t-statistic: $t =$

Poll Concept Check #1

- Which of the following statements are true about hypothesis testing under a simple linear regression?
- I. To find evidence for a negative slope parameter, the null hypothesis is that the slope parameter is less than 0.
- II. A two-tailed t-test will have the same degrees of freedom compared to a one-tailed t-test on the same dataset.
- III. If a t-test fails to reject the null hypothesis that the slope parameter is 0, then there is a relationship between the response and explanatory variables.
- IV. A large p-value leads to rejecting the null hypothesis.

2. t-test Illustration

- You fit the linear regression model: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ to 10 observations to model # of food allergies a person using $x_i = 1$ if the i^{th} person's family member has a food allergy and $x_i = 0$ otherwise.
- 30% had a family member with a food allergy.
- $\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2 = 72$
- $\hat{\beta}_1 = 3$
- Calculate the t-statistic used to test the hypothesis: $H_0\colon \beta_1 = 0$ against $H_a\colon \beta_1 \neq 0$ and determine the smallest listed significance level α for which H_0 is rejected.

2. t-test Illustration 2

- t-ratio $= \frac{Estimator - H_0\ value}{se(Estimator)}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- $MSE = s^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{SSE}{n-2}$

2. *t*-test Illustration 3

- Table

2-sided alpha	20%	10%	5%	2%	1%
df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
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Example 2 (CAS)

- A simple linear regression is run on 15 observations.
- $\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1} = \begin{bmatrix} 0.5 & -0.2 \\ -0.2 & 0.1 \end{bmatrix}$
- $b_1 = 0.7$
- $\sum_{i=1}^{n}(y_i - \bar{y})^2 - b_1 \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = 20$
- Determine the smallest listed α where $H_0: \beta_1 = 0$ is rejected.

Example 2 Sol

- $Var(b) = \begin{bmatrix} Var(b_0) & Cov(b_0, b_1) \\ Cov(b_0, b_1) & Var(b_1) \end{bmatrix} = MSE(\boldsymbol{X}^T\boldsymbol{X})^{-1}$
- $MSE = \frac{SSE}{n-2}$

Confidence Intervals

- A **confidence interval** gives a reasonable range that the true <u>parameter</u> value could fall within, given a significance level α.
- A $100(1-\alpha)\%$ confidence interval in basic linear regression is:
- $b_0 \pm t_{critical} * se(b_0)$
- $b_1 \pm$

Prediction Intervals

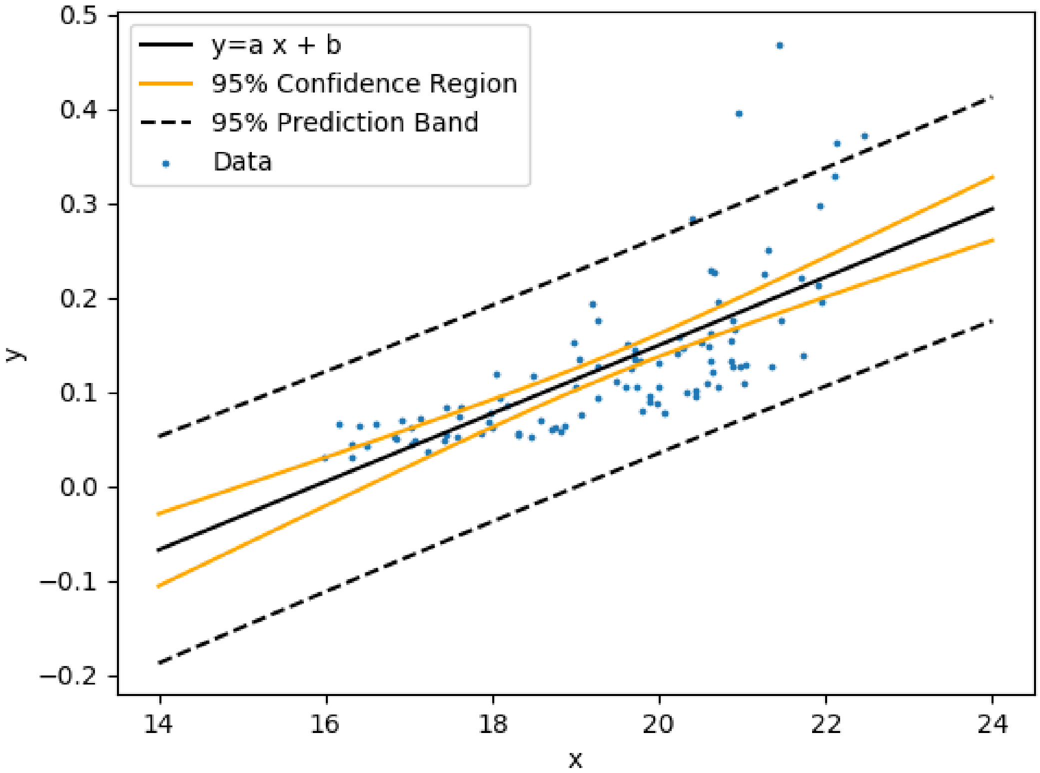
- A **prediction interval** gives a reasonable range for a particular prediction, given a significance level α.
- A $100(1-\alpha)\%$ prediction interval at x_{pred} in basic linear regression is:
- $\hat{y}_{pred} \pm t_{critical} * se(\hat{y}_{pred})$
- The prediction interval has to account for the uncertainty from the ___

Prediction Assumptions

- Predictions assume the new observation ___
- Narrower confidence/prediction intervals are ___

Confidence vs. Prediction

- Both intervals are narrowest at $\bar{X}$.
- Prediction interval is at least as wide as confidence interval (due to irreducible error)



Intervals Summary

- Use confidence intervals for ___

- Use prediction intervals for ___
- If a $100(1-\alpha)\%$ confidence interval for b_i includes 0, then we can conclude that ___

Intervals and α

- The confidence interval and α represent two sides of the same coin.

Confidence Interval	80%	90%	95%	98%	99%
2-sided alpha	20%	10%	5%	2%	1%
df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
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Poll Concept Check #2

- In our basic linear regression model, which statements are true?
- I. If there is no irreducible error, the confidence interval is equal to the prediction interval.
- II. The prediction interval quantifies the possible range of $E[b_1|x]$.
- III. The prediction interval is always at least as wide as the confidence interval.
- IV. The bias of ordinary least squares regression is 0.

3. Intervals Illustration

- A simple linear regression is fit on the following dataset to predict Y (in thousands) based on number of credentials X.
- Use your calculator as much as possible!
- A. Calculate the 95% confidence interval for β_1.
- B. Find the p-value range of the hypothesis test $H_0: \beta_1 = 4$.

Y	50	60	65	70	75	80
X	0	1	3	2	4	4

3. Intervals Illustration 2

- $s^2 = \frac{SSE}{n-2}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}} = \frac{s}{\sqrt{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}}$
- $b_1 \pm t_{critical} * se(b_1)$
- t-ratio $= \frac{Estimator - H_0\ value}{se(Estimator)}$
- Do 2-Var Stats first.
- Data → Formula → Add/Edit Formula
- Predictions: L3 = b (from StatVars) + a*L1
- e^2: L1 (replace x) = (L2 – L3)^2 (will generate recursive error but still work)

3. Intervals Illustration 3

df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
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8	1.3968	1.8595	2.3060	2.8965	3.3554
9	1.3830	1.8331	2.2622	2.8214	3.2498
10	1.3722	1.8125	2.2281	2.7638	3.1693
11	1.3634	1.7959	2.2010	2.7181	3.1058
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123

Example 3 (SOA)

- You fit the model $y_t = \beta_0 + \beta_1 x_t + \epsilon_t$ to 10 observed values (x_t, y_t).
- $\sum(y_t - \hat{y}_t)^2 = 2.79$
- $\sum(x_t - \bar{x})^2 = 180$
- $\sum(y_t - \bar{y})^2 = 152.4$
- $\bar{x} = 6$
- $\bar{y} = 7.78$
- Determine the width of the shortest symmetric 95% prediction interval for y when $x = 8$.

Example 3 Sol

- Prediction Interval: $\hat{y}_{pred} \pm t_{critical} * se(pred)$
- $Var(\hat{y}_{pred}) = MSE\left(1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $MSE = s^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{SSE}{n-2}$

Summary Check

- *Can you describe in your own words:*
- Formulas for standard error of $b_0, b_1, \hat{y}, \hat{y}_{pred}$
- How to conduct a t-test and use the table
- How to find confidence and prediction intervals